

Predicting Online Shopper Purchase Intentions from Session Behaviour

Abstract

The problem of our project is to determine whether it is possible to predict the outcome of a shopping session based on people's behaviour during it. With this, we plan to use the Online Shoppers Purchasing Intention dataset from the UCI Machine Learning Repository, which has 12,330 instances and 17 features, such as time spent on websites, number of pages visited, and the time of year of the visit. The main value we want to predict is revenue, which shows whether the session ended in a purchase. We plan to clean and prepare the data, handle any outliers and missing values, encode necessary information, and then split it into training, validation, and testing sets. Afterwards, we will train a logistic regression model to predict whether a purchase will occur, as well as a decision tree to estimate how likely a purchase is based on browser behaviour. Finally, we will compare both models against a simple baseline and measure their performance using precision, recall, and F-1 score.

Research Background

Although many people browse online retailers' websites, not every customer completes a purchase. Understanding which browsing behaviours and patterns are associated with purchasing decisions can help businesses improve their online websites and customer experience.

Previous research by Sakar et al. (2019) explored the use of machine learning to predict purchase intentions using Random Forest, Support Vector Machines, and neural networks. Their research demonstrates how customer browsing data can be used to predict purchasing outcomes. Our project aims to explore how simpler machine learning models, such as logistic regression and decision trees, can predict whether a customer will complete a purchase.

Methodology

Dataset

We will use the Online Shoppers Purchasing Intention Dataset from the UCI Machine Learning Repository. The dataset contains 12,330 instances with 17 input variables, including time spent browsing, pages visited and period of visit. The output variable, Revenue, indicates whether a purchase was made.

Data Processing

We will examine the dataset, clean and trim out outliers and missing values, encode categorical variables, and scale numerical features. The dataset will be divided into training, testing and validation to ensure our models are evaluated correctly.

Models and Methods

We plan to use two supervised learning methods

- Logistic Regression: Predict whether a session will result in a purchase
- Decision Tree: Predict purchase probabilities using decision rules and boundaries based on browsing behaviours.

Evaluation

We will compare logistic regression and decision trees against a simple baseline using precision, recall, and F1-score.

Significance

This project aims to contribute to how machine learning can be used to predict online purchasing intentions and behaviour. By identifying browsing patterns associated with purchases, these findings will help online retailers understand customer behaviours and refine their marketing strategies.

SMART Timeline

Milestone	Activities	Deliverable
M1 - Fri Sep 18	Finalize proposal and team contract; create the shared repository; all members review the dataset	Proposal PDF + contract + repository link
M2 - Fri Oct 2	Select a dataset, run EDA, define the target, splits, metrics, and risks	Data card + EDA notebook + defined target
M3 - Mon Oct 19	Build the data pipeline, implement the baselines, and make results reproducible	Working pipeline + logistic regression baseline with a results table
M4 - Fri Oct 30	Review methods and results, identify blockers, update timeline	Decision tree added + comparison table + progress slides presentation
M5 - Fri Nov 20	Train and compare both models, analyze errors, and draft conclusions	Compare logistics regression and decision tree model + draft results and conclusions
M6 - Fri Dec 4	Finalize results, document contributions	Finalize results + report + repository + presentation demo
M7 - Mon Dec 7	Each member reflects on their own contribution	Individual reflection PDF

References

“Supervised Machine Learning.” *GeeksforGeeks*, GeeksforGeeks, 9 May 2026, www.geeksforgeeks.org/machine-learning/supervised-machine-learning/.

Sakar, C. & Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5F88Q>.

Sakar, C. Okan, et al. “Real-Time Prediction of Online Shoppers’ Purchasing Intention Using Multilayer Perceptron and LSTM Recurrent Neural Networks - Neural Computing and Applications.” *SpringerLink*, Springer London, 9 May 2018, link.springer.com/article/10.1007/s00521-018-3523-0.